Accident Pattern and Risk Intelligence System in India (2018-2020)

1. Introduction

Road accidents are a major public safety issue in India, leading to a high number of deaths and injuries every year. Understanding accident patterns helps in identifying risk factors and improving road safety measures. This project analyzes accident data from 2018, 2019, and 2020 to study trends, severity, and high-risk categories.

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2. Objective

- To analyze accident data from 2018–2020
- To identify trends in victims and offenders
- To calculate accident severity rate
- To find high-risk states and transport types
- To provide suggestions for improving road safety

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3. Dataset Description

- Sample size: 500 rows (sample dataset)
- Years covered: 2018, 2019, 2020
- Data includes:
    - Number of victims
    - Number of offenders
    - Transport type
    - State-wise accident data

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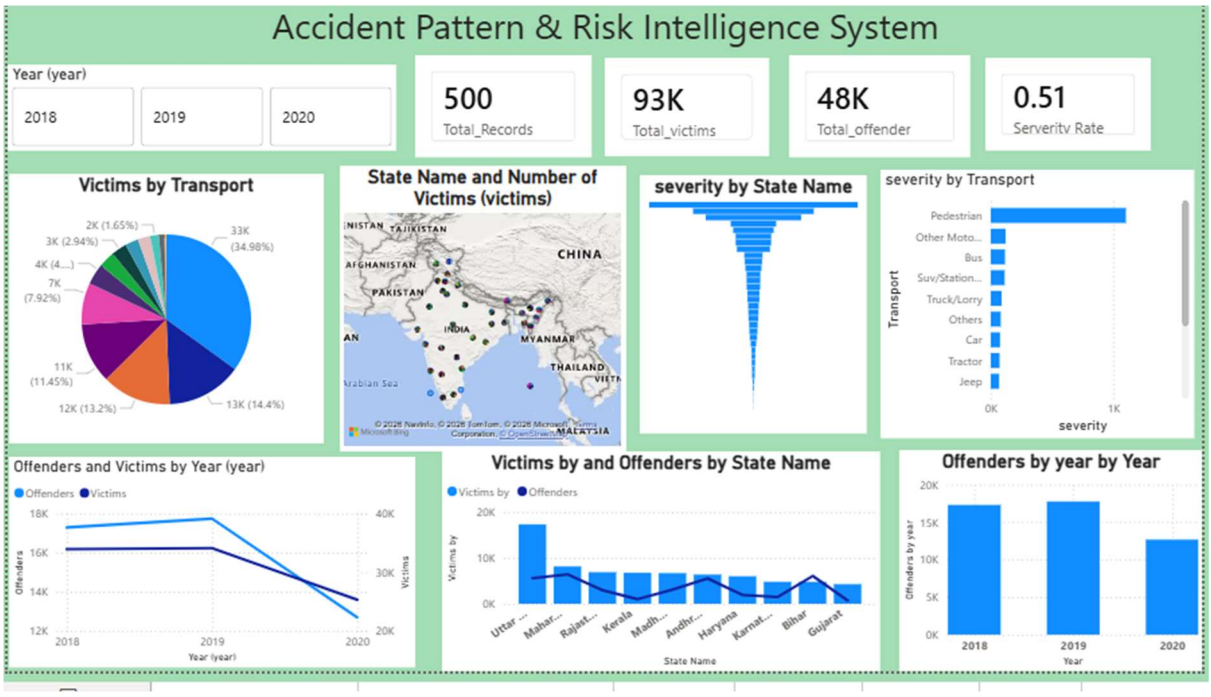
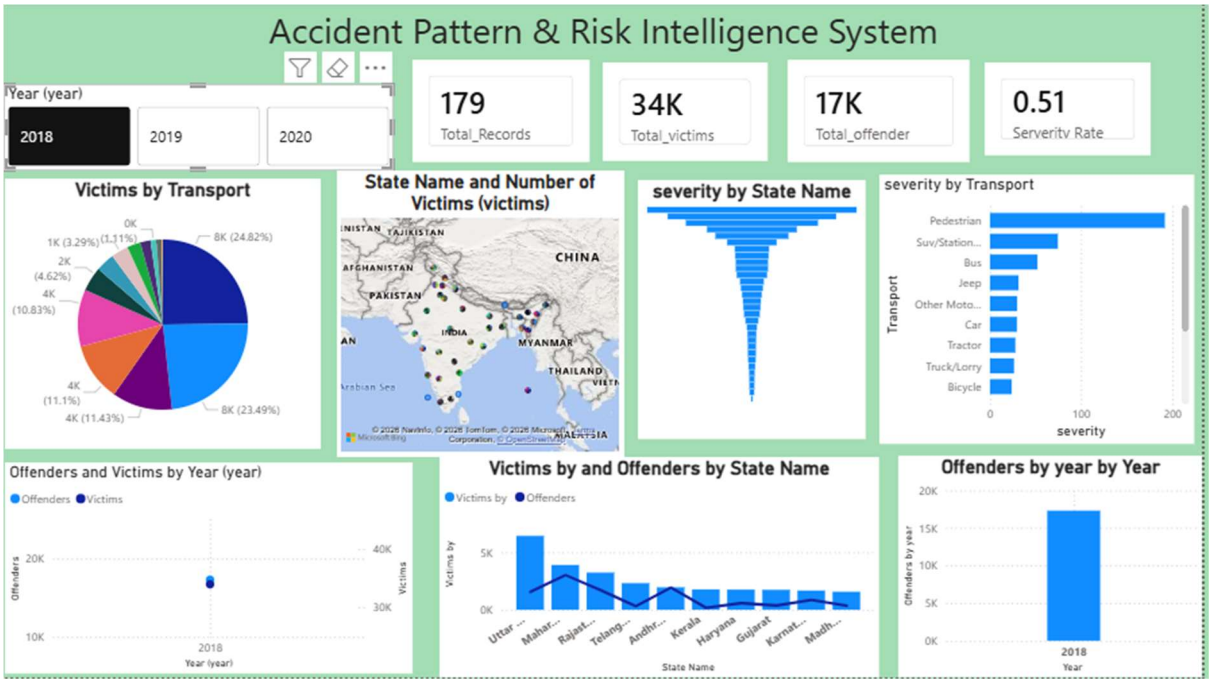
4. Methodology

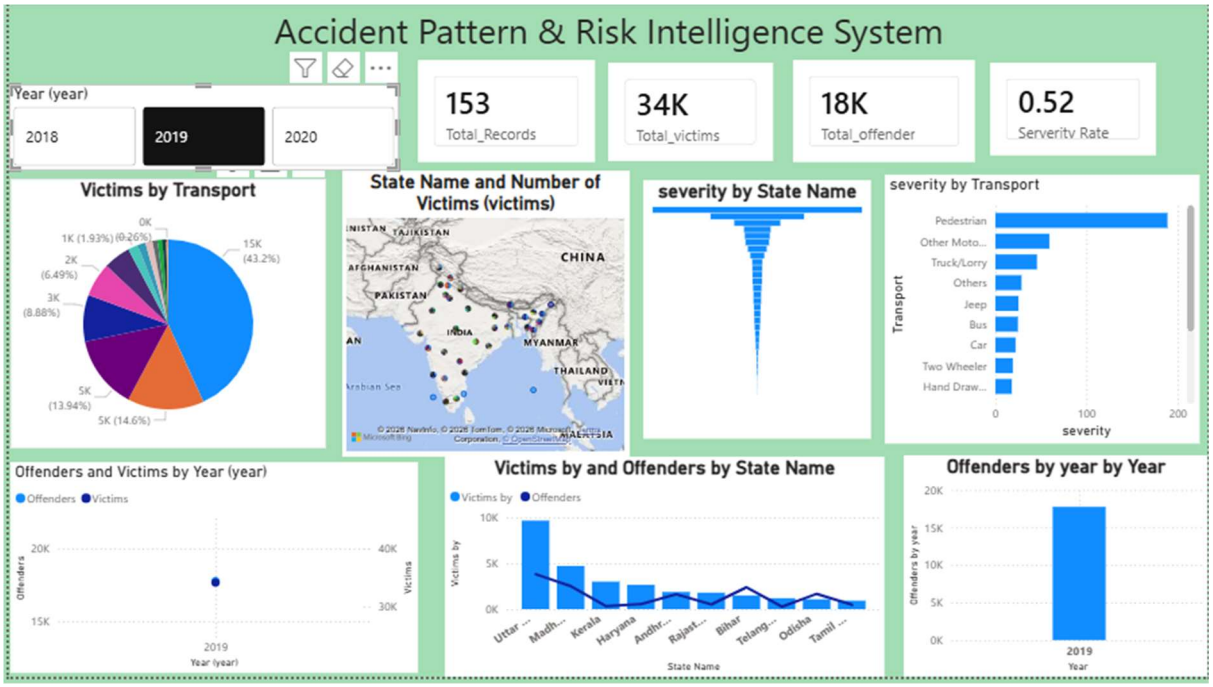
- Data cleaning and preprocessing
- Calculation of severity rate
- Year-wise comparison
- Transport-wise analysis
- State-wise analysis

Severity Rate Formula:

Severity Rate = Number of Victims / Number of Offenders







Types of Analysis (4 Reports)

1.Descriptive Analysis

Key Findings:

- Total victims:
  - 2018 → 34,000

    - 2019 → 24,000
    - 2020 → 24,000
- Total offenders:
    - 2018 → 17,000
    - 2019 → 18,000
    - 2020 → 13,000

Insights:

- Sharp drop in victims from 2018 to 2019
- 2020 remains low (possibly due to lockdown effects)
- Offenders peaked in 2019

2.diagnosisAnalysis

Key Findings:

- Severity Rate:
    - 2018 → 0.51
    - 2019 → 0.52
    - 2020 → 0.50
- Victims by transport:
    - 2018 → Truck (24.82%), Two-wheeler (23.49%)
    - 2019 → Transport category ~43.2% (dominant share)
    - 2020 → Two-wheeler (39.1%)

Insights:

- Severity is consistently high (~50%)
- Two-wheelers and trucks are major contributors
- Pedestrians have the highest severity across all years

☞ Suggests vulnerable road users (pedestrians, two-wheelers) are at highest risk.

3.Prescriptive Analysis

Compare across years and states.

Key Findings:

- Highest severity states:

    - 2018 → Gujarat
    - 2019 → Kerala
    - 2020 → Karnataka
- Transport severity:
    - Pedestrians highest in all 3 years

Insights:

- Severity shifts between states → regional factors matter
- Pedestrian risk is consistently high → major safety concern

☞ Shows trends and changing risk geography.

4. predictive Analysis

Based on your findings, suggest actions.

Risk Intelligence Insights:

- High-risk groups:
    - Pedestrians
    - Two-wheeler riders
- High-risk regions:
    - Gujarat (2018), Kerala (2019), Karnataka (2020)

Recommendations:

- Improve pedestrian infrastructure (footpaths, crossings)
- Enforce helmet laws for two-wheelers
- Target state-specific policies
- Use AI-based accident prediction systems

☞ This turns your analysis into real-world solutions.

Conclusion

- Accident severity remains high across all years
- Pedestrians are the most affected group
- Two-wheelers contribute significantly to accidents
- High-risk states vary each year

This analysis helps in understanding accident trends and identifying critical risk areas.